Quiz # 01 **Course Title:** Peripherals, Interfacing & Embedded Systems **Total Marks:** 15
Course ID: CSE - 4619

Time: 25 Mins. **Student Id: ...……..…** **Obtained Marks: …...**

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Q1. What are the main reasons to use microcontroller? What is the maximum clock speed for ATMega16 and how it can be implemented at ATMega16 microcontroller? **05**

Q2. Suppose, you are given an analog quantization size of 0.75 Volt, where V_{min}=0 Volt and V_{max}=12 Volt. Calculate the desired number of bits for an A/D converter. **05**

Q3. Suppose, it is given V_{in} = 0.75 Volt, V_{ref} = 1 Volt and 8-bit of resolution for a Successive Approximation A/D converter. Find a 8-bit digital output for the given V_{in}. **05**

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